

Reading a Regression Discontinuity Result

Module 2 · Day 2, Session 2: What Happens at the Cut-off

3ie

0 · Where Module 1 left us

What you saw in Module 1

Module 1 compared businesses either side of the eligibility cut-off, in the neighbourhoods where GreenWaste was offered, and reported two numbers.

From memory: why does a comparison at a cut-off count as fair?

Module 1, straight lines -1,119 AED

Module 1, with covariates -905 AED

The rule at least 1,000 AED

Businesses just either side of the line are near-twins. Which side they land on is close to chance, so the line creates the comparison.

Today we open the box

Why did adding covariates move the answer from -1,119 to -905? Today we rebuild the estimate from the data and run the four checks an RDD result has to pass.

- 1 The rule and the data: who was offered, who took part
- 2 From the picture to a number
- 3 Four checks: the window, bending the rule, balance, who it applies to

You will not write code. You call the choices; I type and press **Run**.

1 • The rule and the data

The rule that decided who got in

In the offered neighbourhoods, GreenWaste went to businesses whose **efficiency index** was **at or below 58**: the less efficient ones.

Two businesses either side of that line are almost identical. One got the programme, one did not.

Running variable	Efficiency index
Cut-off	58
Eligible	at or below 58
Rule followed?	we check it next

Read the direction carefully. The rule offers the programme to the **less** efficient businesses, so the eligible side is on the **left** of the chart. Getting this backwards flips the sign of everything that follows.

[MENTI PLACEHOLDER] Predict the picture

Businesses at or below 58 got the programme, and it cut costs. On the chart of costs against the index, **where is the eligible side, and at the line does it sit above or below the other side?**

A Left, lower · **B** Left, higher · **C** Right, lower · **D** Right, higher



Go to **menti.com**, code **1234 5678**, or scan the code.

A. A low index sits on the left, and low is eligible; a programme that cut costs puts the eligible side **below** at the line. Anyone who assumed the programme went to the *better* businesses would read the chart backwards.

Did the rule hold?

R Code

Start Over

Run Code

```
1 data.frame(side = c("above 58", "58 or below"),
2             businesses = as.vector(table(offered$below)),
3             offered = as.vector(tapply(offered$eligible, offered$below, mean)),
4             took_part = as.vector(tapply(offered$enrolled, offered$below, mean)))
```

side	businesses	offered	took_part
above 58	1995	0	0
58 or below	2964	1	1

Before I press Run: what share of eligible businesses do you expect took part?

Every business at or below 58 was offered and took part; none above 58 did. The rule is **sharp**, and the jump at the line is the effect of taking part.

Which businesses belong in the comparison?

- Offered neighbourhoods only
- Every neighbourhood

```
d <- if (sample == "Every neighbourhood") everyone else offered
data.frame(jump_at_the_line = jump_at(d, width = 2)$jump,
           eligible_who_took_part = round(mean(d$enrolled[d$below == 1]), 2))
```

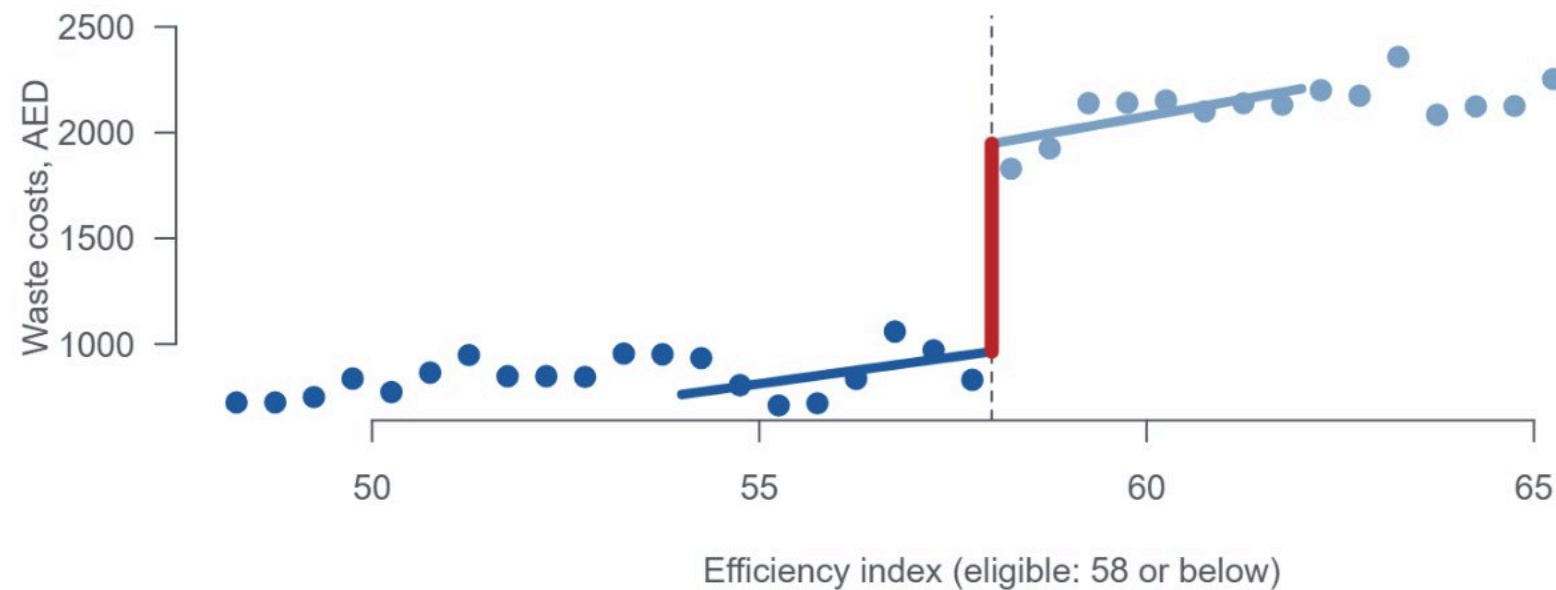
jump_at_the_line	eligible_who_took_part
-791	1

Switch to every neighbourhood. **Why does the jump shrink, and why does take-up fall to about half?**

In the other neighbourhoods an “eligible” business was offered nothing. Mixing them in puts untreated businesses on the eligible side, so the jump falls from about -791 to -249 AED. Ask the evaluator exactly which businesses are in the comparison.

The picture the method is built on

```
jump_plot(width = 4)
```



Dots are average costs in half-point bins. **Is there a step at the dashed line?**

2 · From the picture to a number

The idea behind the method

Compare businesses **just below** the cut-off, who got the programme, with businesses **just above** it, who did not.

Right at the line, scoring 57.9 or 58.1 is close to a matter of chance, so the two sides are alike except for the programme. The **step in costs at the cut-off** is the estimated effect.

A first, rough number

R Code

↺ Start Over

▶ Run Code

```
1 near <- subset(offered, abs(distance) <= 2)      # within 2 points of the line
2 eligible_side <- mean(near$waste_management_costs[near$below == 1])
3 other_side    <- mean(near$waste_management_costs[near$below == 0])
4 round(data.frame(eligible_side, other_side, gap = eligible_side - other_side))
```

eligible_side	other_side	gap
938	2034	-1097

Each pair: call out the jump on your card. **Why are they so different?** Then Run: the same subtraction on every business within 2 points.

Six businesses a side is too few to trust. And even with many, an average across the window mixes in the slope of costs along the index. The regression deals with both.

Running the regression

R Code

↺ Start Over

▶ Run Code

```
1 near <- subset(offered, abs(distance) <= 2)
2 fit <- lm_robust(waste_management_costs ~ below * distance, data = near)
3 round(as.data.frame(summary(fit)$coefficients[, c(1, 2, 4)]), 1)
```

	Estimate	Std. Error	Pr(> t)
(Intercept)	1768.3	108.6	0.0
below	-790.9	149.3	0.0
distance	237.3	98.0	0.0
below:distance	-199.4	127.1	0.1

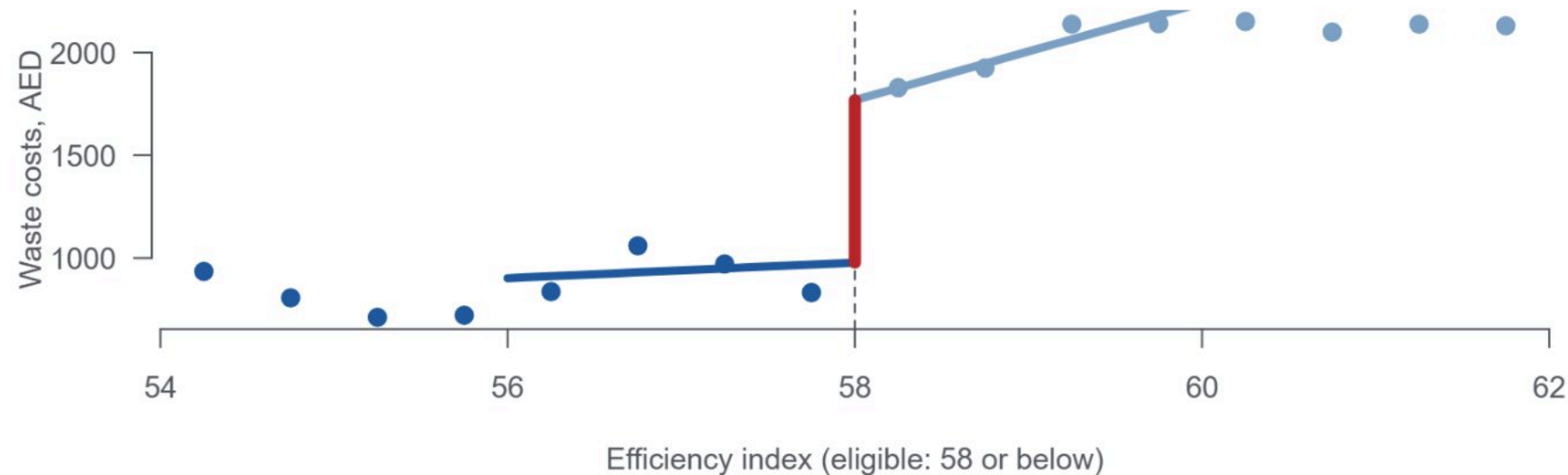
`distance` is the score measured from the line (0 = 58); `below` is 1 on the eligible side. `below * distance` fits a line on each side.

Only one row is the programme's effect. Which one?

below: the jump, -791 AED. `distance`, about +237 AED per point, is the slope **along** the line, not the step **at** it.

Seeing the jump: the red line is below

```
jump_plot(width = 2, view = 4)
```



A line on each side, fitted to businesses within 2 points of 58. The red segment at the line is the **below** row, -791 AED. The lines slope up, which is why the raw average overstated the gap.

Does it clear the rule?

-791

AED at the cut-off, within 2 points

The rule **1,000** Least it could be **498** Clears the bar **No**

A real effect, clearly not zero. The interval runs from 498 to 1,084. **Would you scale the programme on this?**

3 · Four checks

This method's one assumption

Nothing else jumps at the cut-off.

A business one point below 58 and one point above should be alike in every way except whether it got the programme. If something else also changes at 58, the jump is not only the programme.

This morning's difference-in-differences assumed the groups moved in parallel; this method assumes nothing else changes at the line. **If something else did change at 58, where in the data would you see it?**

The GreenWaste lab: add RDD

Joiners differ in something
nobody measured (AED)



Joiners' costs were already falling
faster (AED)



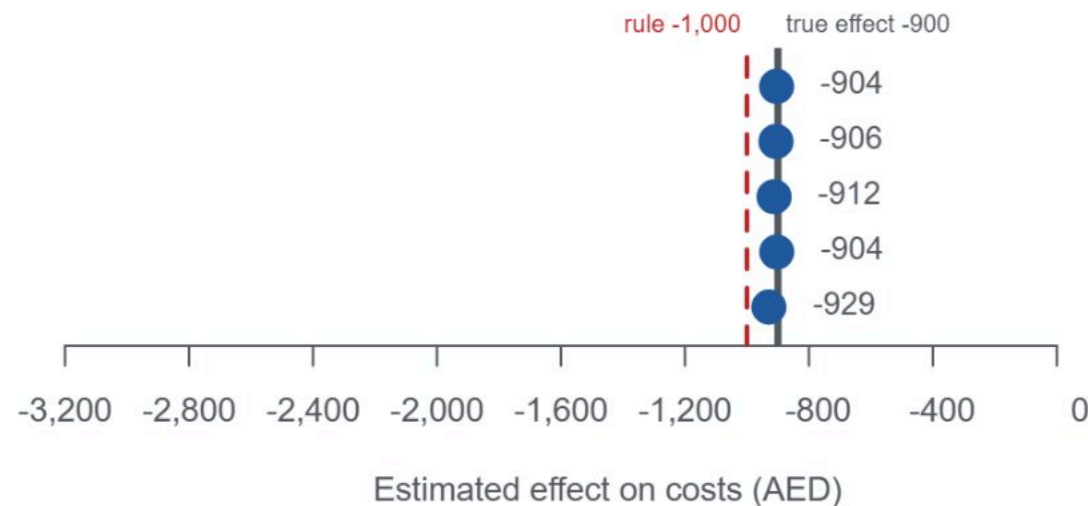
Something else changes at 58
(AED)



RDD joins the lab, with one new problem: the last slider. **Which slider breaks RDD, and which of DiD's problems does it survive?**

Only a jump at 58 breaks it. Differences that change smoothly along the index leave the step at the line alone.

Before and after
With and without
Randomised
Difference-in-differences
Regression discontinuity



Check 1: does the jump survive a different window?

Window either side of 58 (index points)



`jump_at(offered, width)`

cut_off	window	businesses	jump	lower	upper
58	2	773	-791	-1084	-498

The window is the analyst's choice. Slide from 1 to 10. **Does the verdict against 1,000 AED change?**

The estimate moves between about 790 and 1,120 AED, above the bar from a window of 5. The least generous end of the interval never reaches 1,000 (at most about 960). Ask which window was used, and why.

Check 1, continued: a jump where there is no rule?

Pretend the cut-off is at

62



```
jump_at(offered, width = 2, cut = placebo)
```

cut_off	window	businesses	jump	lower	upper
62	2	658	-121	-491	248

Move the cut-off to 50, 54, 62, 66, where no rule exists. **What should the jump be?**

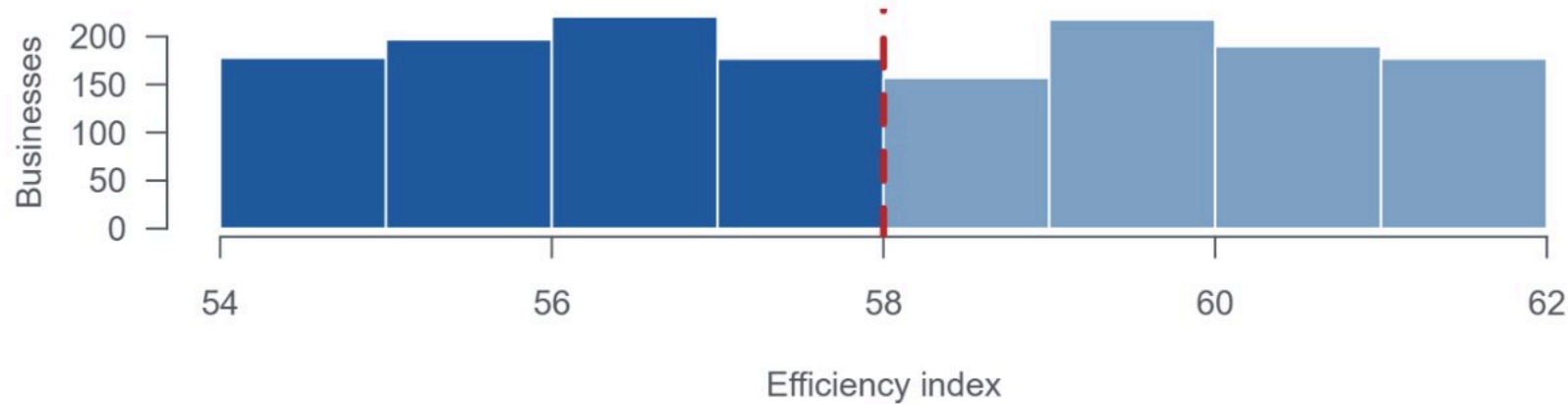
Close to zero, and every interval there contains zero. At 58 the jump is far outside that range. If the method found jumps where there is no programme, its jump at 58 would mean nothing.

Check 2: did anyone bend the rule?

R Code [Start Over](#)

[Run Code](#)

```
1 density_plot()
```



If businesses could nudge their score to 58 or below to qualify, what would the bar just left of the line look like?

Taller than its neighbours, with a dip just right. The bars at the line are not out of pattern: no sign of a pile-up. Ask whether anyone drew this chart; a pile-up only shows if someone looks.

Check 3: are the two sides alike?

Window (index points)



`balance(bwidth)`

characteristic	jump	lower	upper	differs
manager_age	-5.52	-10.44	-0.59	TRUE
staff	0.59	-0.07	1.24	FALSE
area	-0.30	-1.12	0.52	FALSE
recycling_distance	-5.74	-18.24	6.76	FALSE

Each row runs the same regression with a business characteristic as the outcome. If the sides are alike, no row should jump. Change the window. **Which row does not behave?**

Manager age: businesses just below the line have managers 5 to 9 years younger, at every window. Something other than the programme changes at 58. This check fails.

What that row means

If younger managers also run their waste costs differently, part of the jump is the managers, not the programme. That is why Module 1's estimate with covariates (-905) differs from the one without (-1,119).

Adjusting for manager age only fixes what was measured. If manager age jumps at 58, something unmeasured may jump too.

What would you ask the evaluator? *Why does manager age change at the cut-off, and what else might?*

Check 4: who does the result apply to?

R Code

↺ Start Over

▶ Run Code

```
1 near <- abs(offered$distance) <= 2
2 data.frame(group = c("within 2 points of 58", "everyone else"),
3             businesses = c(sum(near), sum(!near)),
4             staff = round(c(mean(offered$staff_size[near]), mean(offered$staff_size[!near])), 1),
5             area = round(c(mean(offered$business_area[near]), mean(offered$business_area[!near])), 2))
```

group	businesses	staff	area
within 2 points of 58	773	4.8	1.76
everyone else	4186	5.3	2.10

The estimate describes the businesses near the line: somewhat smaller, with smaller premises. It does not describe the businesses far from it.

If the plan is to extend the programme to everyone, what have you learned about the businesses the extension would reach?

4 • Put AI to work

AI Snapshot: who does this apply to?

Paste the regression table and the four checks into an AI tool with this prompt:

“Explain, without jargon, who this result actually applies to.”

Mark every claim the study supports.
Four minutes.

A response like this is possible, especially with a short or leading prompt:

This study found that the GreenWaste programme reduced costs by around **790 AED** for participating businesses. **Because the study uses a cut-off at an efficiency index of 58, the findings apply to all businesses in the programme. The cut-off design means the result generalises to the wider population of inefficient businesses.** The estimate is therefore a reasonable basis for **scaling the programme to every business below average efficiency.**

Debrief: one error, three times

1 · “Applies to all businesses in the programme.” The estimate is local to businesses near the line.

2 · “Generalises to the wider population.” Backwards: RDD uses only the businesses closest to the line, so it generalises least.

3 · “A basis for scaling to every business.” A scale-up needs what happens away from the cut-off, which this design discards.

And it never mentions the manager-age jump. The question it was asked, *who does this apply to?*, is the one it answered least well.

The same question, a better prompt

What people actually type

Explain, without jargon, who this result actually applies to.

Nothing about the design, so the model may reach for the most natural-sounding story.

What to type instead

This is a regression discontinuity result with the cut-off at an efficiency index of 58, estimated within 2 points either side. The businesses near the line are smaller than average, and manager age differs across the line. Name which businesses the estimate describes and what it cannot tell us. **Ask me questions before you answer.**

Run both. Does error 1 survive? What did each answer explain well? The habit: give the model the constraint it cannot infer, then ask a narrower question.

5 · Close

Your turn: ask AI first

Before we show our list, ask an AI tool: **“I am a commissioner about to hear an evaluator present a regression discontinuity result. What are the key questions I should ask?”**

Three minutes, then compare with the next slide.

Five questions to ask any evaluator presenting RDD

- 1 Where is the cut-off, which side is eligible, and exactly which units are compared?
- 2 Does the jump survive other windows, and is there none at placebo cut-offs?
- 3 Is the number of units smooth at the line, or did anyone bend the rule?
- 4 Does anything else jump at the line, such as manager age?
- 5 Who is near the line, and does the estimate clear our rule for them?

Where this sits

- Day 1 S3** naive comparisons and a randomised trial: -1,014, interval spans 1,000
- Day 2 S1** difference-in-differences: -816 against a 1,000 rule
- Today** regression discontinuity: -791 near the line (-905 with covariates), against 1,000

DiD gets its comparison by assuming parallel trends. RDD gets it from the rule, and the estimate describes only the businesses near the line. Next session, matching covers more businesses by assuming everything that matters was measured.